TOWARDS A BRAIN COMPUTER INTERFACE USING WAVELET TRANSFORM WITH AVERAGED AND TIME SEGMENTED ADAPTED WAVELETS

Muhammad Bilal Khalid, Naveed Iqbal Rao, Intisar Rizwan-i-Haque, Sarmad Munir, Farhan Tahir
Telecommunications Engineering Dept., MCS, National University of Sciences and Technology (NUST)
Rawalpindi, Pakistan
{bilalkhalid, naveedi, intisar, sarmad, farhan} @mcs.edu.pk

Abstract—**A Brain Computer Interface (BCI) is a system that provides an individual a non-muscular way to control his surroundings by working on the activity signals generated in his brain. Activity signals collected from the brain in the form Electroencephalogram (EEG) are used in the system described in this paper. Time Domain normalization has been used in the preprocessing stage. Wavelet transform with pattern adapted wavelet filter is used for the purpose of feature extraction. Exact Radial Basis Neural networks are used for the classification of signals and heuristic weight adjustment of neural networks was done in final decision making. Tests carried out on two different datasets give accuracies of more than 91% and 86%.**

Keywords-Brain Computer Interface (BCI); Wavelet transform; Pattern adapted wavelet

I. INTRODUCTION

A BCI system comprises of five main parts; signal acquisition, preprocessing (signal enhancement), feature extraction, classification and control interface blocks. Signal acquisition system directly captures EEG or any other signals from the brain. It also performs noise elimination, artifact processing etc. The aim of preprocessing block is to bring the signals into a suitable form for further processing. Feature extraction portion extracts features from the signals, forming feature vectors, upon which classification can be done. The classification block classifies the signals using their feature vectors and control interface translates the classified signals into meaningful commands for any device connected, such as a wheel chair or a computer.

This work proposes a novel approach for the second, third and fourth blocks of a BCI system. Non-linear time domain normalization for the second block has been proposed. Other techniques commonly used are Common Average Referencing (CAR), Surface Laplacian (SL), Independent Component Analysis (ICA), Common Spatial Patterns (CSP), and Principal Component Analysis (PCA) [18]. Neural networks perform well when their input signal range is limited. We do this by using non-linear normalization used by [1], but in time domain. Time domain normalization requires much less computing power and provides for faster processing which is a critical factor in online implementation of BCI systems. Current trends in feature extraction indicate usage of Fast Fourier Transform (FFT) [1], [14], Parametric modeling and spectral parameters [2], Signal envelope cross correlation [3]. In this work we propose using an adapted averaged wavelet filter for transformation. This adapted wavelet depicts EEG signal characteristics, such as drift, trends, abrupt changes, and beginnings and ends of events, much better and hence improve feature extraction. Wavelet transform has also been used by [4] and others [18], but this and other techniques that use wavelet transform, use a predefined wavelet from some wavelet family like Haar, Meyer, Morlet and Daubechies wavelets, although pattern adapted wavelets have been used [6], [7], [9] and others [18], but the difference between this method and previous ones is that this wavelet is adapted not only to the individual tasks under consideration but also their specific time segments, thus helping create a very efficient filter. Heuristic weight adjustment of the Neural network classifiers has also been proposed. This weight adjustment, as described in subsequent sections, provides for a much better classification accuracy by utilizing the vital information of accuracy of individual electrodes feeding the networks.

To summarize, this work has three contributions; first in preprocessing we have used non-linear normalization in the time domain. Secondly wavelet transform with pattern adapted wavelet filter because it provides better resolution in time-frequency plane, thirdly heuristic weight adjustment of neural network classifiers. The selection of exact radial basis network for classification provided an improved efficiency in terms of time and memory usage of the system.

II. PREPROCESSING

A. Data

First dataset (dataset1) used was acquired from [13]. This dataset also been used by [1], [12], [14] and many others. Seven Non-Invasive active electrodes C3, C4, P3, P4, O1, O2 and EOG were used to record EEG signals at a sampling frequency of 250Hz from seven individuals, six males and one

Sponsored by: Higher Education Commission of Pakistan and Image Processing Center, MCS, NUST, Pakistan

female. Therefore for a 10 sec recording 250×10 = 2500 samples were recorded. The Dataset comprises of 5 Mental Tasks and each task was repeated five times on two days, amounting to total trials of 50. These Tasks were; Baseline task, Multiplication task, Letter Composing task, 3-D Rotation task and Counting task. The second dataset (dataset2) was obtained from [17]. This dataset was recorded from a normal subject (female, 25y) during a feedback session. The subject sat in a relaxing chair with armrests. The task was to control a feedback bar by means of imagery left or right hand movements. The order of left and right cues was random. The experiment consists of 7 runs with 40 trials each. All runs were conducted on the same day with several minutes break in between. Given are 280 trials of 9s length. The first 2s was quite, at t=2s an acoustic stimulus indicates the beginning of the trial, the trigger channel (#4) went from low to high, and a cross “+” was displayed for 1s; then at t=3s, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of the cue. The feedback was based on Adaptive Autoregressive (AAR) parameters of channel #1 (C3) and #3 (C4), the AAR parameters were combined with a discriminant analysis into one output parameter. The recording was made using a G.tec amplifier and Ag/AgCl electrodes. Three bipolar EEG channels (anterior ‘+’, posterior ‘-‘) were measured over C3, Cz and C4. The EEG was sampled with 128 Hz, it was filtered between 0.5 and 30Hz.

B. Windowing

To reduce the execution time and memory utilization in processing 10secs and 9secs EEG signal for dataset1 and dataset2 respectively, each signal was segmented into windows of 1sec with an overlap of 0.5sec. Overlap was introduced so as to cater for the loss at the edges of the window and to make BCI response fast as each new window was available after 0.5sec. This segmentation does not affect any cross validation effort, because this much overlap does not make signals identical to each other or dependent on each other over times greater than the windowing time.

C. Time Domain Normalization

Neural Networks when used for classification perform far better if the input data lies in a small range of values. Often important information is present in small samples which could be overwhelmed by large insignificant sample values if not appropriately scaled. For that purpose the data was non-linearly normalized using a normalization technique proposed by [5]. Mathematically

$$y = \frac{\log(x - x_{\min})}{\log(x_{\max} - x_{\min})} \tag{1}$$

Where in (1) y is the normalized sample, x is the current sample, and x_{max} and x_{min} are maximum and minimum values in the 1 sec window under consideration. It must be remembered that this normalization was done in time domain, prior to feature extraction, as opposed to [1], which proposes normalization of the FFT coefficients in frequency domain. Data from dataset1 before and after normalization is shown in Fig. 1. Horizontal axis shows samples and vertical axis shows the magnitudes. The difference between blue (before normalization) and red (after) shows how the samples (on the horizontal axis) are now limited to a small range of magnitude (on the vertical axis).

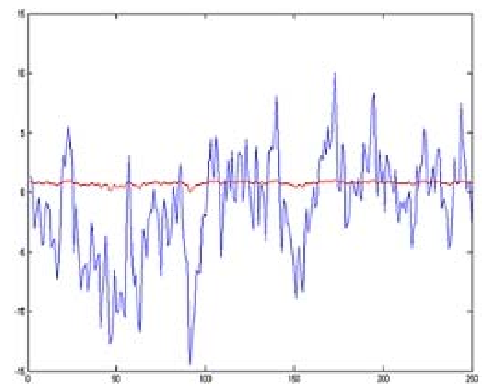

Figure 1. Data before (blue) and after (red) normalization

III. FEATURE EXTRACTION

This section describes the logic behind the selection of wavelet analysis as feature extraction tool. In transforming to the frequency domain using Fourier transform, time information is lost. When looking at a Fourier transform of a signal, it is impossible to tell when a particular event took place. If the signal is a stationary signal, this drawback isn't very important. However the EEG signals contain numerous non stationary or transitory characteristics: drift, trends, abrupt changes, and beginnings and ends of events. Wavelet analysis provides a much better resolution, that is, it allows the use of long time intervals where we want more precise low-frequency information, and shorter regions where we want high frequency information. The major advantage afforded by wavelets is the ability to perform local analysis, that is, to analyze a localized area of a larger signal. Wavelet Transform is a representation of a signal x(t) by a basis function Ψ having such properties as smoothness, localization of Time Frequency, orthogonality and symmetry obtained by scaling and shifting the mother wavelet. Mathematically;

$$X(\alpha, \beta) = \int_{-\infty}^{\infty} x(t)\psi(\alpha, \beta, t)dt \tag{2}$$

where α and β are real numbers representing scale and position of the wavelet respectively.

The appropriate selection of basis function Ψ, the wavelet, is significant as it should approximate a given pattern in the signal. By merely looking at the brain signals one might erroneously consider them to be highly random in nature, which in fact they are not. There is a pattern associated with every task that human eye cannot decipher. The fundamental assumption in pattern detection is that data is not random and that the pattern repeats itself with a slight variation where pre-processing is required to remove this variation to the extent possible and to make the pattern detectable. For that very reason use of wavelet transform was made. The basic idea behind using an adapted wavelet is to use a wavelet for analysis which actually depicts the signal characteristics and thus provides for better feature extraction.

Pattern adapted wavelet using least Square Optimization was selected for pattern detection of these signals. A wavelet was adapted for each window segment of data using polynomial method and degree of polynomial was set to be five. Wavelet for each task in both datasets was obtained for one second window using Matlab® Wavelet toolbox command ‘pat2cwav’, i.e. ‘pattern to continuous wave’. This command computes an admissible wavelet for CWT adapted to the given pattern. The principle for designing a new wavelet for CWT is to approximate a given pattern using least squares optimization under constraints leading to an admissible wavelet well suited for the pattern detection using the continuous wavelet transform [16]. An average wavelet filter was obtained using simple averaging of the adapted wavelets over all the tasks for both datasets. Wavelets for each window segment, i.e., for each one second window was used in the averaging processes to yield a filter for each of the window segments in the signal, to be used for CWT of that window segment of the training and test data. Thus the CWT coefficients were obtained using adapted wavelet filter for each time segmented window of data. Entire process of synthesizing the filter was carried out with Matlab® wavelet toolbox [8].

Another work, [6], also describes a similar filter for performing wavelet transformation. However the filter described in [6] is adapted to a single neural activity. The filter proposed in this paper takes it one step ahead by providing and testing a wavelet filter prepared by averaging the adapted wavelets of all the tasks under consideration for wavelet analysis. This filter is thus optimized for the tasks under consideration.

This wavelet adaptation scheme combined with data segmentation and normalization yields excellent feature vectors for classification, the reason for it can be explained as; when we make overlapping segments of the given signal, we get a continuous flow of pattern in our resultant segmented signal. Even though signal is time segmented, yet due to their overlapping nature each segment picks up the pattern where the previous one dropped it, thus when we extract wavelets from this pattern, we get extremely useful pattern information in these wavelets.

IV. CLASSIFICATION

A Multilayer Exact Radial Basis (RBE) Neural Network was used for the purpose of mental task classification. An RBE network contains one hidden radial basis layer of as many neurons as input vectors and an output linear layer and has a property of returning the exact target vectors if training and testing data is same. It has an added advantage that its training time is noticeably less compared to other neural networks. Based on the hypothesis that different parts of the brain generates different EEG signals for a task, each electrode data was fed to a different network and parallel structure of neural network was used for classification. One RBE Neural Network was trained for each electrode for both datasets. A parallel structure of networks has been used for classification. This parallel structure provides for a much better classification because each network works on a separate electrode thus fully utilizing the montage characteristics of individual electrodes.

V. HEURISTIC WEIGHT ADJUSTMENT FOR FINAL DECISION MAKING

Decision Making involves weighted voting, that is, taking decisions from all the networks and making final decision about the task selected based on simple majority. During training and initial testing phase weight adjustment was done heuristically i.e. by observing the accuracies of networks (and corresponding electrodes), which provides a means to give more weight to networks with higher accuracy of correct classification in the final decision making. A task was classified as one if majority of the networks classified it to be that particular task. The concept of weight adjustment comes in when we assign weights to networks while deciding. Since the different networks correspond to different electrodes which are feeding them and for which they are trained. Electrodes were found to exhibit different fidelities in representing the different tasks, and based on these differences weight adjustment was made, so as to provide the network (or the electrode) with better accuracy, a greater share in classification.

VI. RESULTS

Weights are assigned to each network (electrode) on basis of results (accuracy) obtained during initial training and testing phase. Accuracy of 91% for dataset1 for all its subjects (averaged) and 86% for dataset2 was achieved during tests. Accuracy can be improved, theoretically to 100%, if intensive iteration is used to continuously monitor and update the weights of the networks. This can be tested by running the algorithm on the same training data, which was done in this work to achieve an accuracy of 99.2% on training data, another work [15] attempting classification on training data reports 98% accuracy. The accuracies achieved, i.e. 91% and 86%, with new test data compare reasonably well with other reported accuracies such as by [1], [10], [11], [15] and others that mostly lay between 60-90%.

VII. CONCLUSION

Time domain normalization of overlapping segmented data provides for a limited range of input values for the neural network classifiers and a ‘smooth’ continuity of pattern. The continuity of pattern when combined with wavelet adaptation for each segment provides excellent feature vectors. A real time (online) implementation of this scheme will benefit from its better response time: since each overlapping time segment is of one second only, all processing can be done fast enough for a real time system. Heuristic weight adjustment of the networks also provides a dynamic way to enhance system performance, e.g. a BCI system may be designed which on user request or after a specified time period runs this weight adjustment algorithm to enhance performance in real time.

REFERENCES

[1] K. Nakayama, Y. Kaneda, A. Hirano, "A Brain Computer Interface based on FFT and Multilayer Neural Network-Feature Extraction and Generalization", Proceedings of 2007 International Symposium on Intelligent Signal Processing and Communication Systems, Xiamen, China, pp. 826-829, 2007.

[2] D. P. Burke, S. P. Kelly, P. de Chazal, R. B. Reilly and C. Finucane, "A parametric feature extraction and classification strategy for brain computer interfacing" IEEE Transactions Neural System Rehabilitation, vol. 13, pp. 12-17, 2005.

[3] Y. Wang, Z. Zhang, X. Gao and S. Gao, "Lead selection for SSVEP-based brain–computer interface", Engineering in Medicine and Biology Society, 2004, IEMBS '04 26th Annual International Conference of the IEEE, vol. 2, San Francisco CA USA, pp. 4507- 4510, 2004.

[4] S. Darvishi and A. Al-Ani, "Brain-Computer Interface Analysis using Continuous Wavelet transform and Adaptive Neuro-Fuzzy Classifier", Proceedings of the 29th Annual International Conference of the IEEE EMBS Cité Internationale, Lyon, France, pp. 3220-3223, 2007.

[5] Khalid, M.B.; Rao, N.I.; Rizwan-i-Haque, I.; Munir, S.; Tahir, F., "A Brain Computer Interface (BCI) using Fractional Fourier Transform with Time Domain normalization and heuristic weight adjustment**",** Signal Processing, 2008. ICSP 2008. 9th International Conference. pp. 2734 – 2737, 26-29 Oct. 2008.

[6] E. L. Glassman, "A Wavelet-Like Filter Based on Neuron Action Potentials for Analysis of Human Scalp Electroencephalographs" IEEE Transactions on Biomedical Engineering, vol. 52, issue no. 11, pp. 1851-1862, 2005.

[7] V. J. Samar, K. P. Swartz, and M. R. Raghuveer, "Multiresolution analysis of event-related potentials by wavelet decomposition", *Brain and Cognition*, Academic Press, vol. 27, pp. 398–438, 1995.

[8] Matlab Wavelet Toolbox. MathWorks.

[9] J. O. Chapa and M. R. Raghuveer, "Optimal matched wavelet construction and its application to image pattern recognition," in Wavelet Applications: II—Proceedings of Society Photo Instrumentation Engineering SPIE, vol. 2491, pp. 518–529,1995.

[10] C. Brunner, R. Scherer, B. Graimann, G. Supp, and G. Pfurtscheller, "Online Control of a Brain-Computer Interface Using Phase Synchronization", IEEE Transactions on Biomedical Engineering, Vol. 53, No. 12, pp. 2501-2506, 2006.

[11] N. F. Ince, A. H. Tewfik, S. Arica, "A Space-Time-Frequency analysis approach for the classification motor imagery EEG recordings in a Brain Computer Interface task", Proceedings of the 28th IEEE EMBS Annual International Conference, New York City, USA, pp. 2581-2584, 2006.

[12] J. B. Ochoa, "EEG Signal Classification for Brain Computer Iterface Applications", Ecole Polytechnique Federale De Lusanne, 2002.

[13] Z. A. Keirn. "Alternative modes of communication between an and machine". Master's thesis, Purdue University, 1988. Colorado State University: http://www.cs.colostate.edu/eeg/

[14] K. Nakayama, K. Inagaki. "A Brain Computer Interface Based in Neural Network with Efficient Pre-Processing". International Symposium on Intelligent Signal Processing and Communication Systems, Tokyo, Japan, pp. 673-676, 2006.

[15] B. Liu, J. Gao, Z. Liu, Z. Zhang, C. Yin, C. Peng and J Gu, "Brainwave Classification based on Wavelet Entropy and Eventrelated Desynchronization". 20th Canadian Conference on Electrical and Computer Engineering, pp. 1108-1021, 2007.

[16] M. Misiti, Y. Misiti, G. Oppenheim, J.M. Poggi, "Les ondelettes et leurs applications", Hermes, 2003.

[17] Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology Graz. (Gert Pfurtscheller).

[18] A. Bashashati, M. Fatourechi, R. K. Ward and G. E. Birch, "A survey of signal processing algorithms in brain–computer interfaces based on electrical brain signals", Journal of Neural Engineering vol. 4, IOP Publishing, pp. 32-57, 2007.